EXPLOITING VULNERABILITIES IN TRADITIONAL AI SYSTEMS AND RISK MITIGATIONS

ATTACKS AND DEFENSE

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Today's focus

Security of AI

- Secure the underlying AI training data by protecting it from sensitive data theft, manipulation, and compliance violations
- Secure model development by scanning for vulnerabilities in the pipeline, hardening integrations, and enforcing policies and access
- Secure the usage of AI models by detecting data or prompt leakage, and alerting on evasion, poisoning, and extraction attacks

AI for Security

- AI will manage repetitive security tasks, such as summarizing alerts and log analysis, freeing teams to tackle strategic problems
- AI will generate security content, expedite implementation, and adjust to changing security threats in real-time
- AI will learn and create active responses, with the ability to find all similar incidents, update affected systems, and patch vulnerable code

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Overview of AI systems

An artificial intelligence system (AI system) is defined as: A machine-based system designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions/recognitions, content, recommendations, or decisions that can influence physical or virtual environments.

Quoted from EU Artificial Intelligence Act, https://artificialintelligenceact.eu/article/3/

Traditional AI systems
(today's class)

Generative/agentic AI systems
(tomorrow's class)

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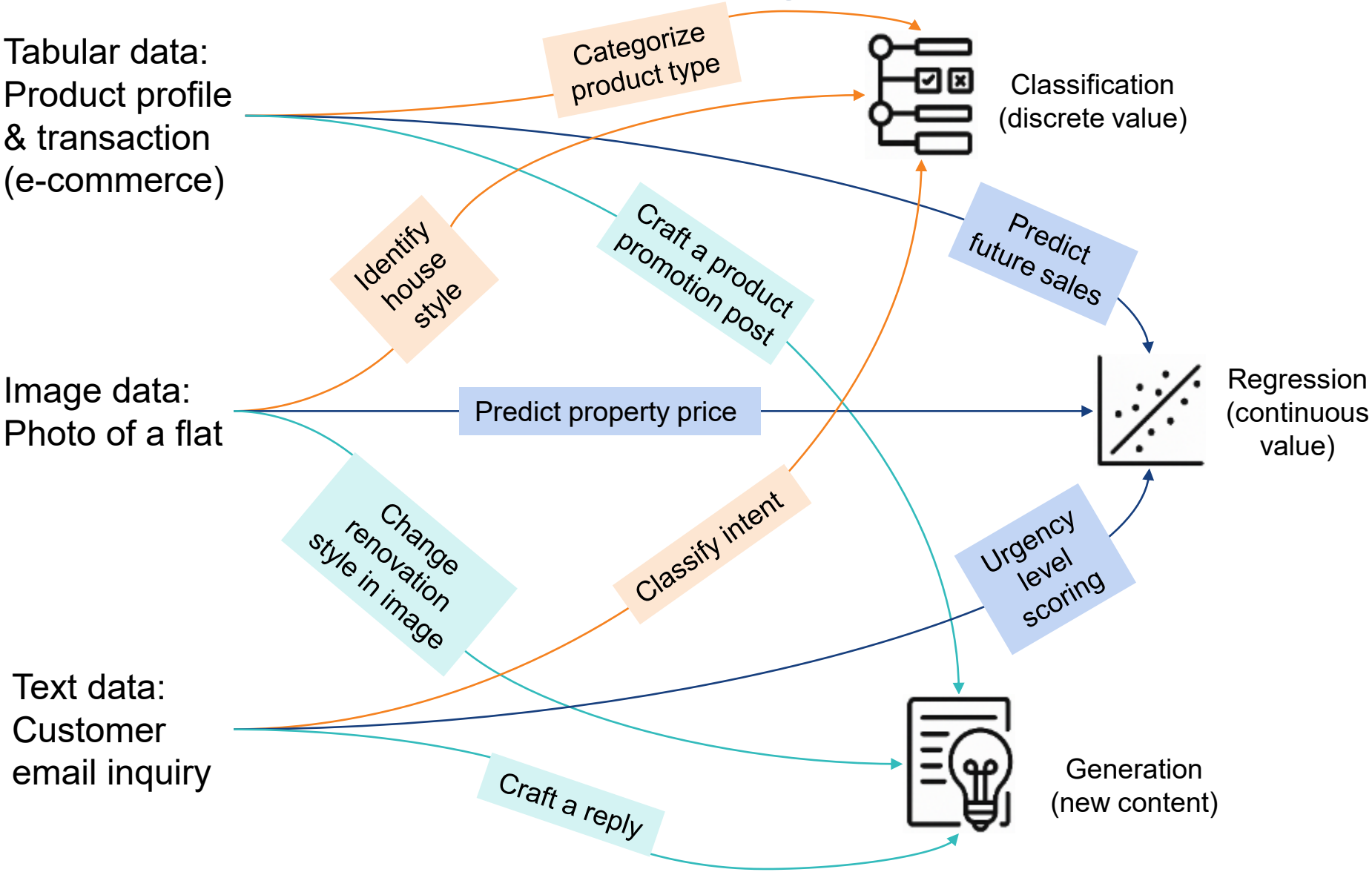
Overview of traditional AI systems: Tasks of machine learning



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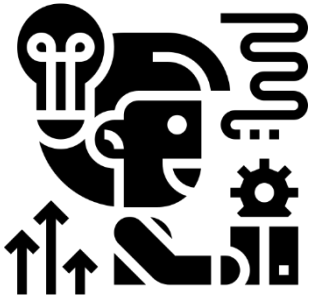
Overview of traditional AI systems: Impacts of machine learning



Automate Repetitive Tasks

Recognition

Video analytics	Video analytics for Kerbside Loading Bay Pilot (URA)
Information extraction and summarisation	Extraction of unstructured financial data (DOS)

Conversation systems

Chatbot	Municipal Services Chatbot (MSO)
Speech and language tools	Transcription of parliamentary sitting (Parliament)

Augment Human Capabilities

Personalisation

Personalised services	Customisation of job recommendations (WSG)

Pattern and Anomaly Detection

Sentiment analytics	Understanding SME issues (ESG)
Feedback analytics	Enabling holistic urban planning (URA)
Fraud analytics	Detection of Fraudulent Claims (SSG)

Forecasting and Decision Support

Predictive modelling	Identification of riskier workplaces (MOM)

Source: https://www.developer.tech.gov.sg/products/collections/data-science-and-artificial-intelligence/playbooks/public-sector-ai-playbook.pdf

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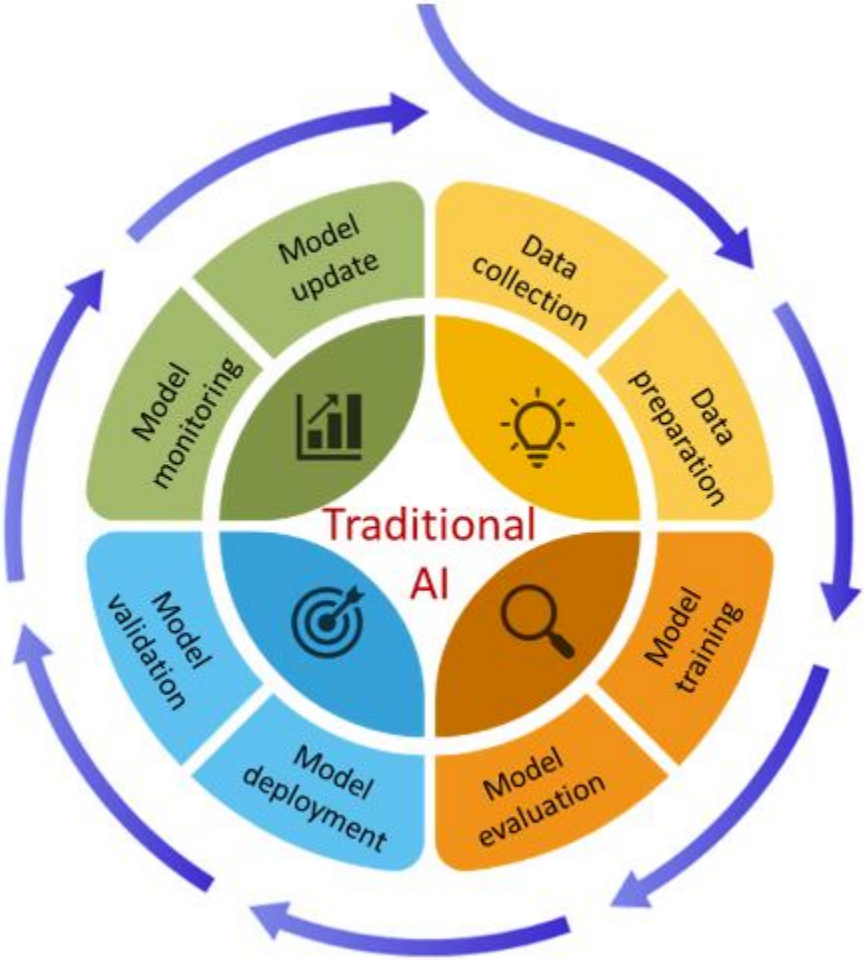
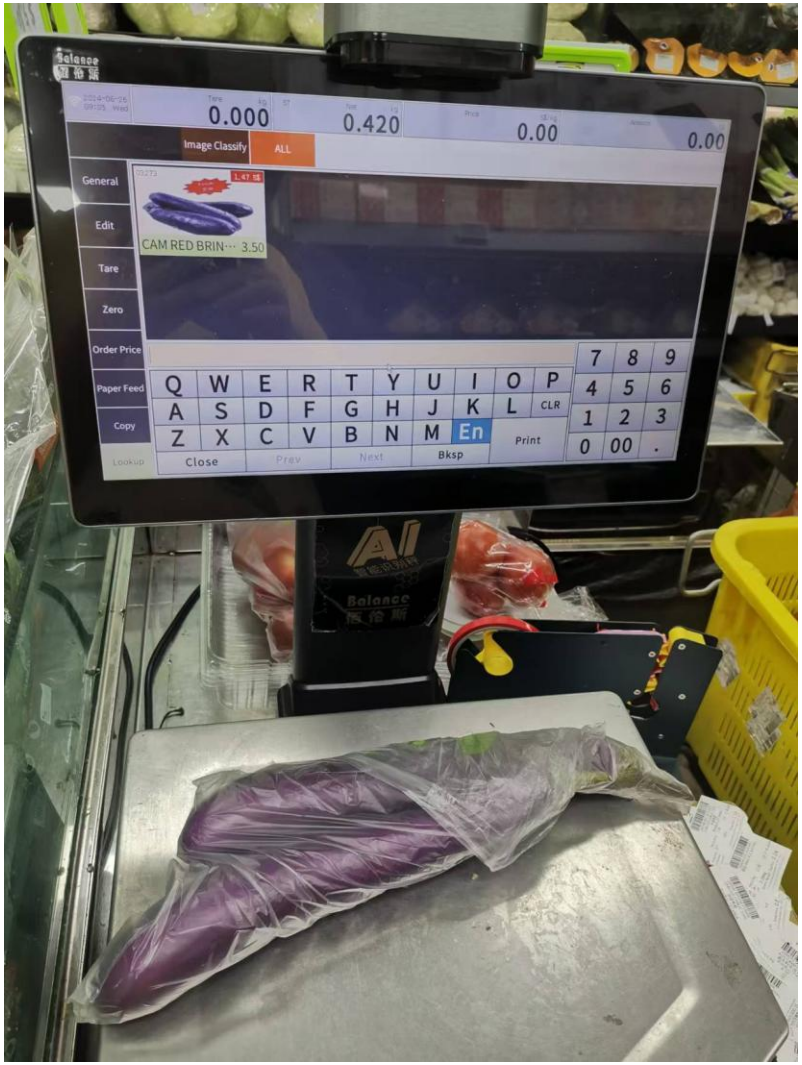
Case study

Traditional AI-powered automatic fruit image recognition at supermarket checkouts





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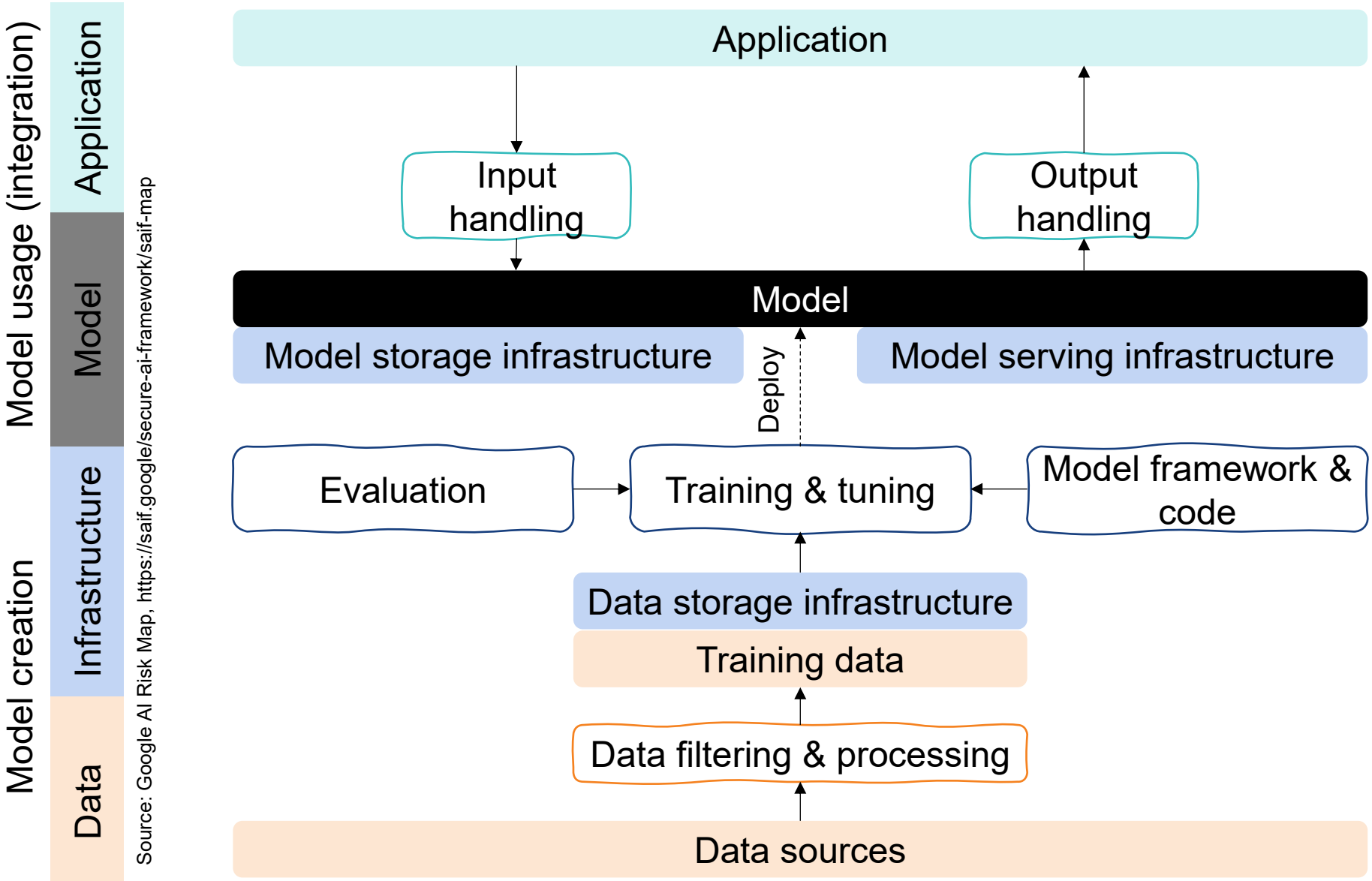
Case study



Source: Google AI Risk Map, https://saif.google/secure-ai-framework/saif-map

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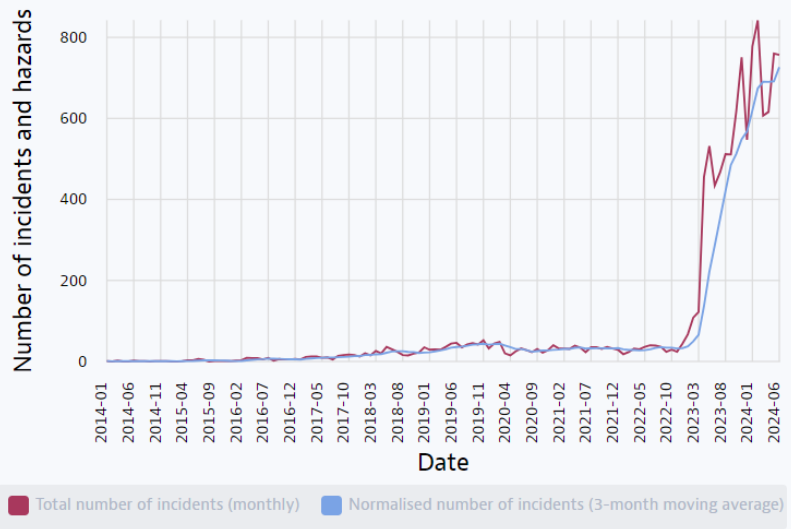
Traditional AI is not robust (vulnerable) (1)

OECD AI Incidents Monitor

Website: https://oecd.ai/en/incidents

Summary visualisations of AI incidents & hazards

Evolution of incidents and hazards



Summary statistics of AI incidents & hazards

	Incidents & hazards	Articles
All time total	11591	64190
Current month's total	10	23
Last month's total	757	3695
Peak month	2024-02	2024-01
Peak amount	842	4486
% change (month-over-month)	-0.39	-12.02
% change (quarter-over-quarter)	-4.22	-4.52
% change (year over year)	305.69	289.66

*Note: Percent change is calculated based on preceding full months (i.e. the current month is excluded).

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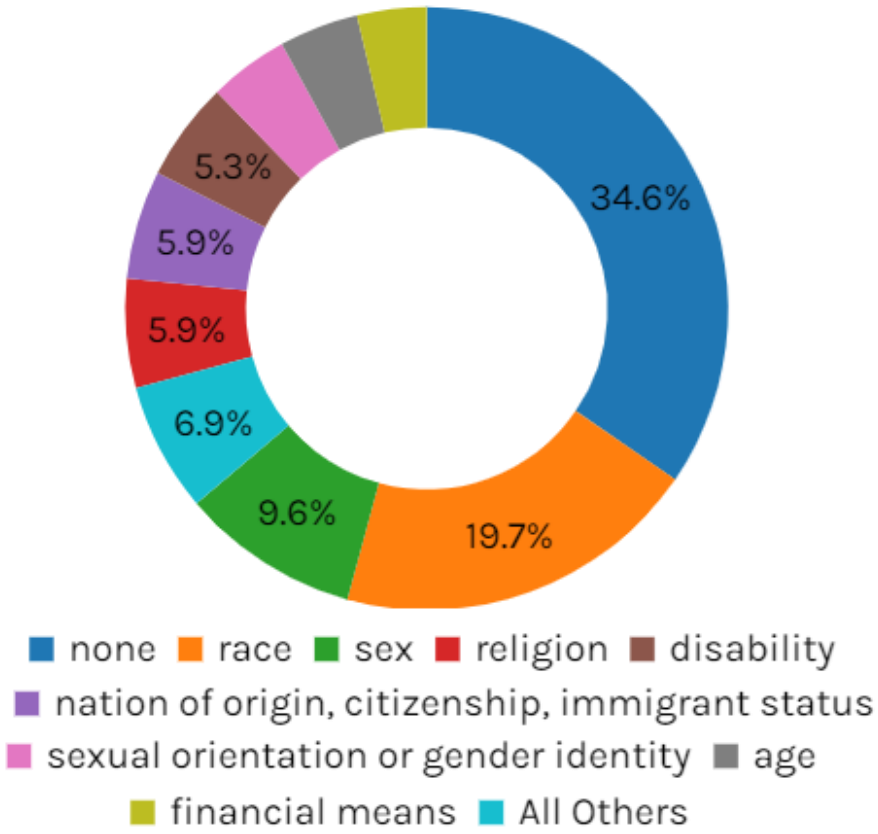
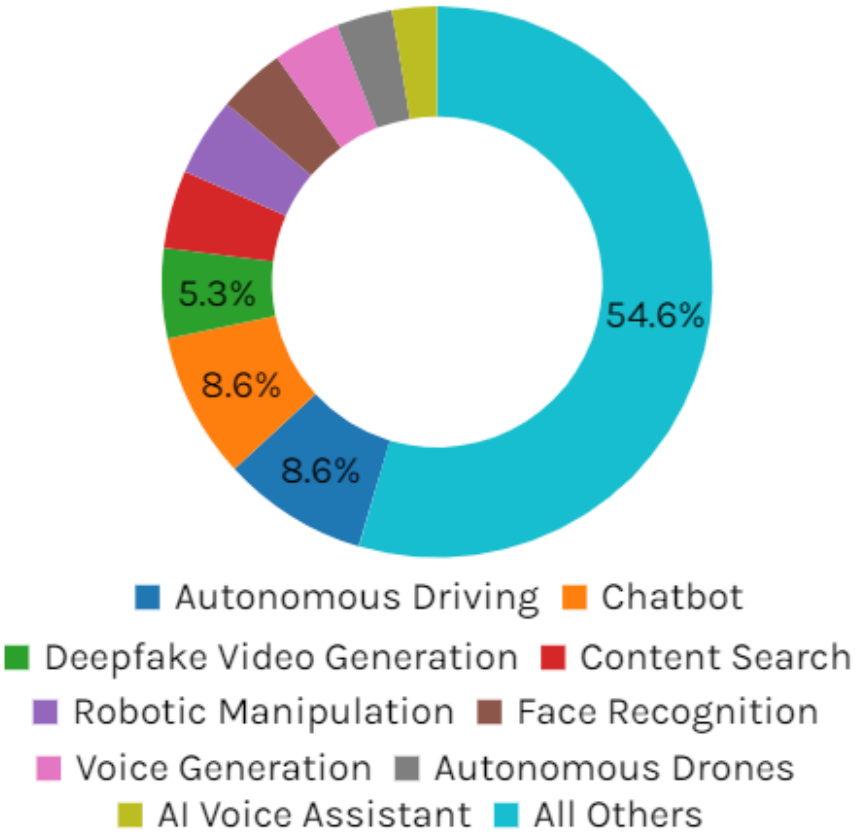
Traditional AI is not robust (vulnerable) (2)

AI Incident Database

Website: https://incidentdatabase.ai/





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Traditional AI is not robust (vulnerable) (3)

Finance

Surveillance

Smart system

- Inaccurate identity verification
- Fake document verification
- Fake voice authentication
- Fraud detection failure

Science News

How secure are voice authentication systems really?

Attackers can break voice authentication with up to 99 per cent success within six tries

Date: June 27, 2023

Source: University of Waterloo

Deepfake Software Fools Voice Authentication With 99% Success Rate

Creating a fake voice to trick authentication systems has never been so easy or effective.

Reference: https://www.sciencedaily.com/releases/2023/06/230627123411.htm;
https://www.pcmag.com/news/deepfake-software-fools-voice-authentication-with-99-success-rate

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Traditional AI is not robust (vulnerable) (4)

Finance

Surveillance

Smart system

- Vulnerabilities of face recognition
- Issues of speech recognition
- Incorrect face analytics

Computer Vision Dazzle Camouflage

CV Dazzle explores how fashion can be used as camouflage from face-detection technology, the first step in automated face recognition. It is a concept and strategy, not a pattern or product, and it is always designed relative to a specific algorithm and unique to each face.

Reference: https://cvdazzle.com/

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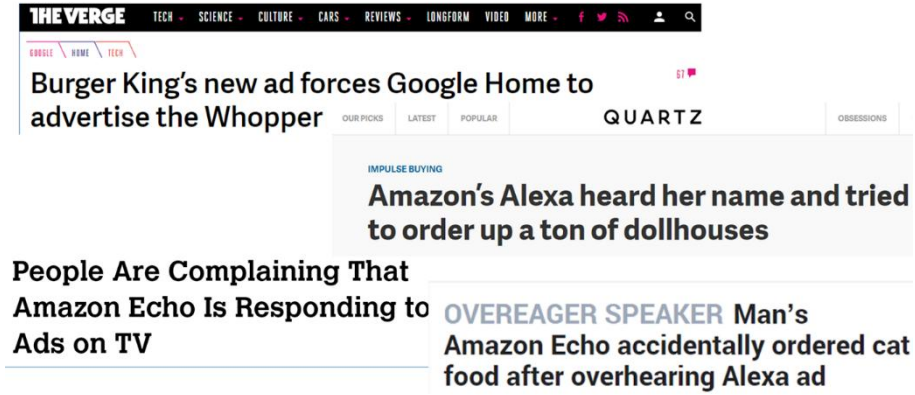
Traditional AI is not robust (vulnerable) (5)

Finance

Surveillance

Smart system

- Identity theft
- Smart system control



Reference: https://secml.github.io/class5/

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Traditional AI is not robust (vulnerable) (6)

Physical Adversarial Attack

Four teenagers arrested after neighbourhood in Punggol vandalised

cna

Vulgarities were seen scrawled at a car park at Block 325 Sumang Walk, with red paint also seen on various signs around the estate.

Reference: Physical adversarial attack meets computer vision: A decade survey, IEEE Trans. on Pattern Analysis and Machine Intelligence, 2024, https://arxiv.org/abs/2209.15179

Photo: https://www.channelnewsasia.com/singapore/four-teenagers-arrested-sumang-walk-punggol-carpark-vandalism-vulgarities-3069661

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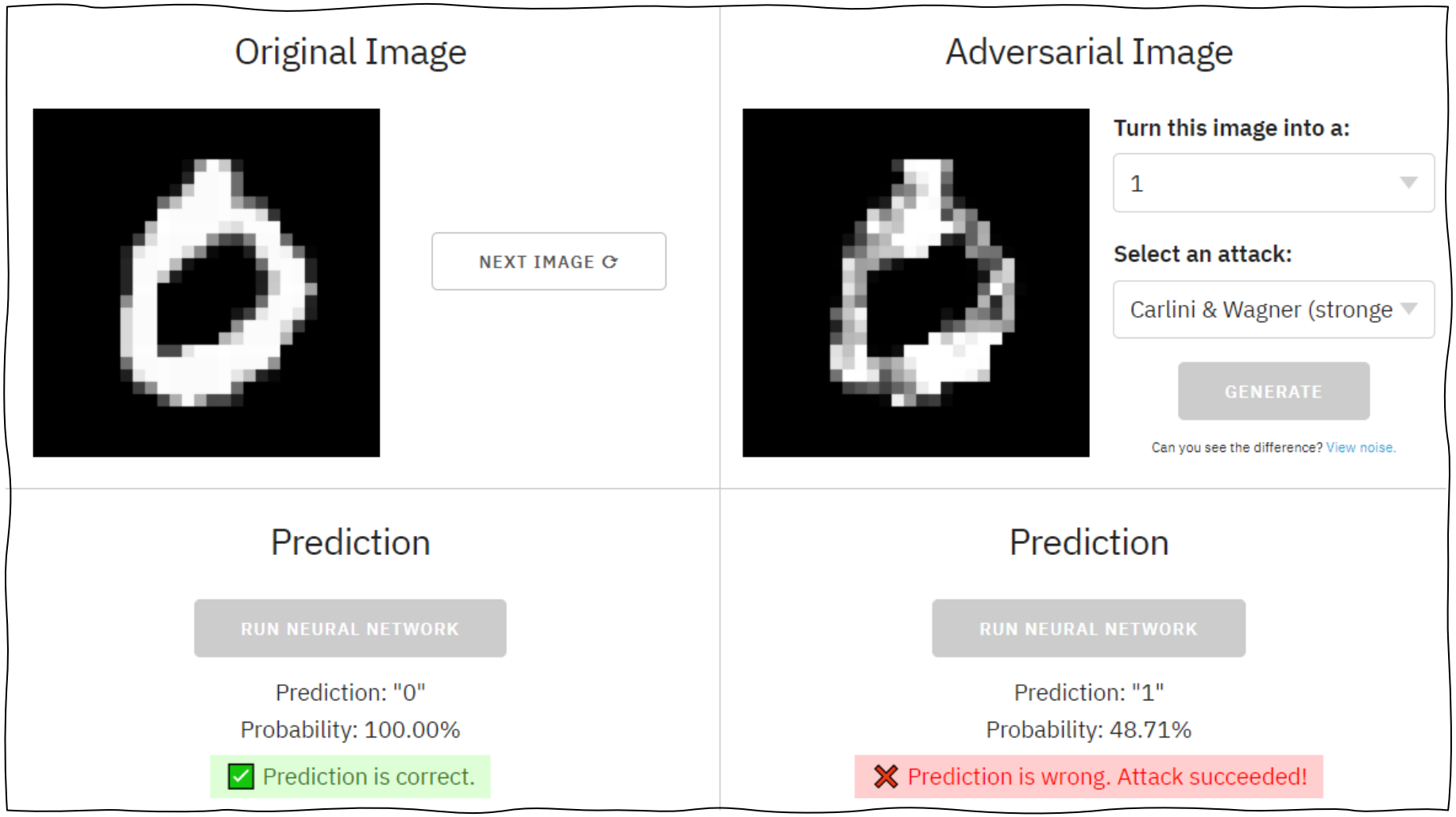
Traditional AI is not robust (vulnerable) (7)

Try it online yourself: https://kennysong.github.io/adversarial.js/



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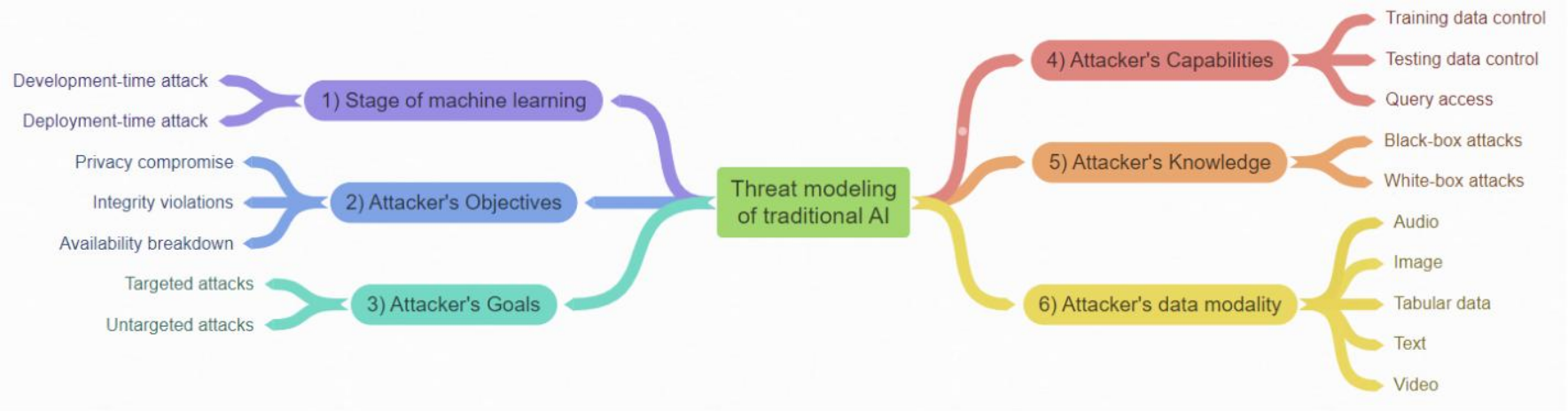
Taxonomy of threat models of traditional AI systems



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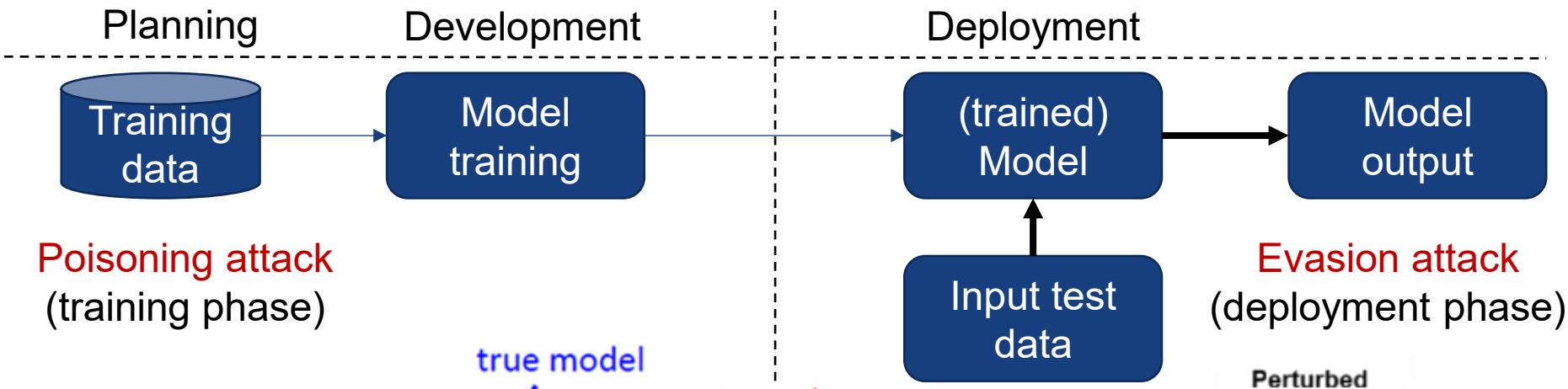
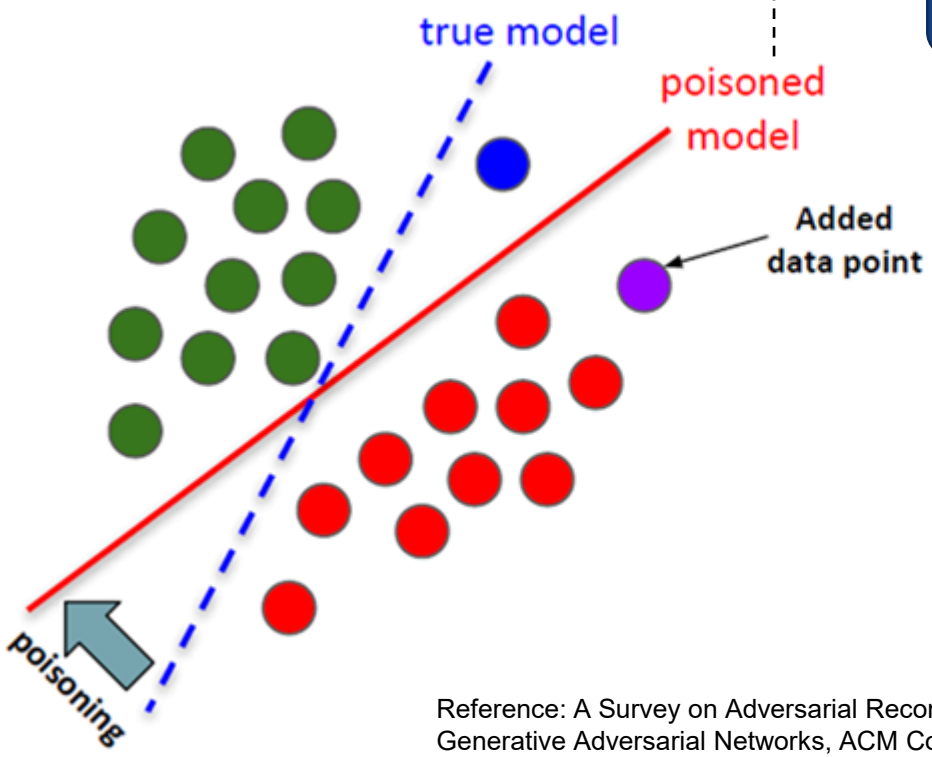
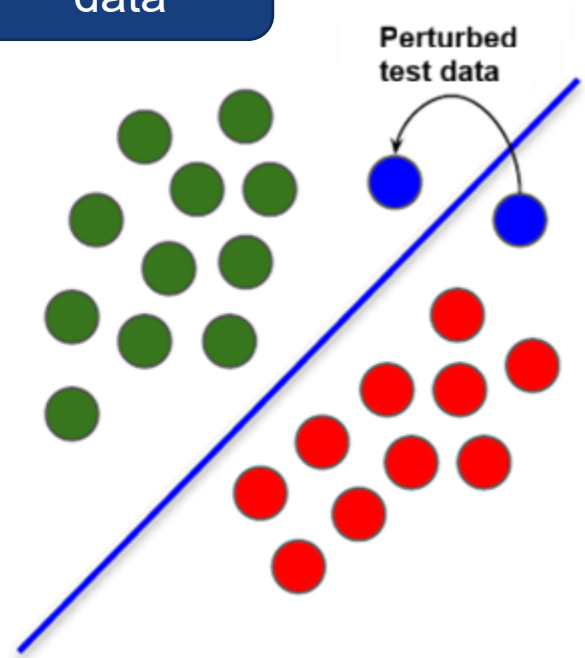
Taxonomy: Stage of machine learning







Reference: A Survey on Adversarial Recommender Systems: From Attack Defense Strategies to Generative Adversarial Networks, ACM Computing Surveys, 2020, https://dl.acm.org/doi/10.1145/3439729

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Taxonomy: Attacker's objectives

- Availability breakdown: An availability attack is an indiscriminate attack against ML in which the attacker attempts to break down the performance of the model at deployment time.
- Integrity violations: An integrity attack targets the integrity of an ML model's output, resulting in incorrect predictions performed by an ML model.
- Privacy compromise: Attackers might be interested in learning information about the training data (resulting in data privacy attacks) or about the ML model (resulting in model privacy attacks).

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Taxonomy: Attacker's goals (classification)

- Targeted mis-classification: The adversary tries to craft the inputs in such a way that the model produces the output of a particular target class. For example, any input image to the classification model will be predicted as a class of images having the 'go' sign.
- Untargeted mis-classification: The adversary tries to alter the output classification of an input example to some other class. For example, a legitimate image of a 'stop' sign will be predicted as any other class different from the class of a stop sign.

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Taxonomy: Attacker's goals (regression)

Change the predicated values, such as Estimation of Arrival time (ETE)

Input	0	Speed	continuous
	1	Flight Distance	
	2	Departure Delay	
	3	Initial ETE	
	4	Latitude Origin	
	5	Longitude Origin	
	6	Altitude Origin	
	7	Latitude Destination	
	8	Longitude Destination	
	9	Altitude Destination	
	10	Arrival Time Slot	7 slots (categorical)
	11	Departure Time Slot	7 slots (categorical)
	12	Aircraft Category	6 classes (categorical)
	13	Airline Company	19 classes (categorical)
Output	3	Refinement ETE	continuous

Reference: An Adversarial Attacker for Neural Networks in Regression Problems, IJCAI 2021, https://centralesupelec.hal.science/hal-03527640v2

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Taxonomy: Attacker's capabilities

- Training data control: The attacker might take control of a subset of the training data by inserting or modifying training samples.
- Test data control: The attacker may utilize this to add perturbations to testing samples at model deployment time, as performed in evasion attacks to generate adversarial examples.
- Query access (API call): When a cloud provider manages the AI model, the attacker might submit queries to the model and receive predictions (either labels or model confidences).

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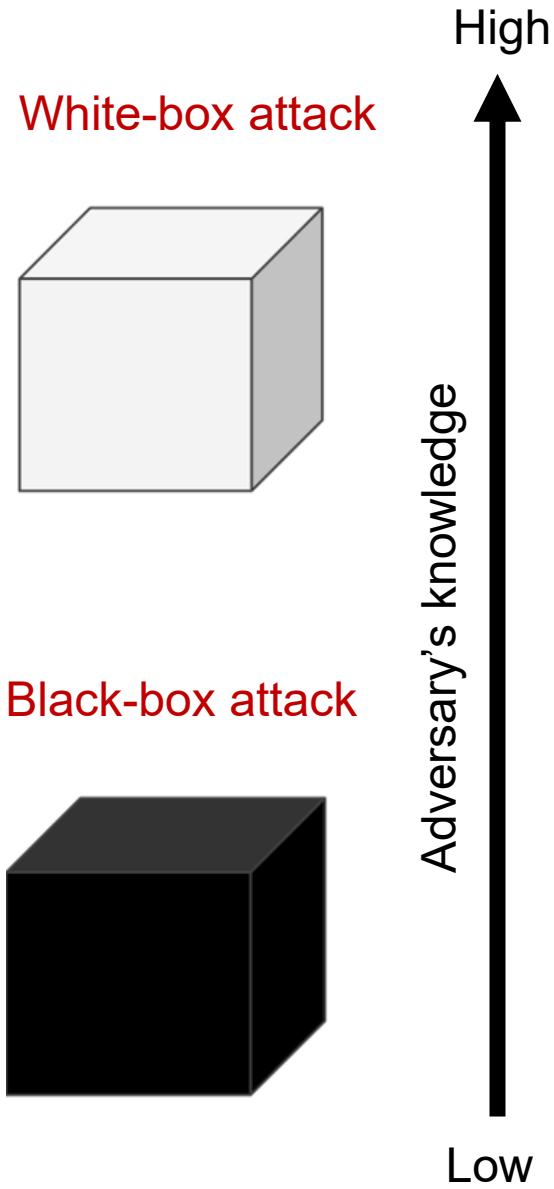
Taxonomy: Attacker's knowledge



- Raw training data and annotation (if any)
- Machine learning model hyperparameters (the secret recipe to train the model)
- Machine learning model parameters (open-weight vs. closed-source)
- Machine learning model architecture (e.g., CNN, transformer)
- Machine learning model outputs (decision and/or confidence scores)

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Taxonomy: Attacker's data modality

- Audio
- Image
- Sensor data (IoT)
- Tabular data (business transaction data)
- Text
- Video

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Case study: Facial biometric systems

Andrew Ng

→ Face detection → → Face recognition →

$$\begin{Bmatrix} 0.01 \\ 0.97 \\ 0.02 \end{Bmatrix} \begin{matrix} \text{Li Fei–Fei} \\ \text{Andrew Ng} \\ \text{Hinton} \end{matrix}$$

Targeted attack: I want to break into Hinton's office
- Targeting a specific subject
- To access specific resources

Untargeted attack: I don't want to be recognized as Ng in an AI conference
- Being recognized incorrectly
- To achieve privacy

$$\begin{Bmatrix} 0.03 \\ 0.05 \\ 0.92 \end{Bmatrix} \begin{matrix} \text{Li Fei–Fei} \\ \text{Andrew Ng} \\ \text{Hinton} \end{matrix} \quad \begin{Bmatrix} 0.35 \\ 0.05 \\ 0.60 \end{Bmatrix} \begin{matrix} \text{Li Fei–Fei} \\ \text{Andrew Ng} \\ \text{Hinton} \end{matrix} \quad \begin{Bmatrix} 0.45 \\ 0.15 \\ 0.40 \end{Bmatrix} \begin{matrix} \text{Li Fei–Fei} \\ \text{Andrew Ng} \\ \text{Hinton} \end{matrix}$$

or other predictions

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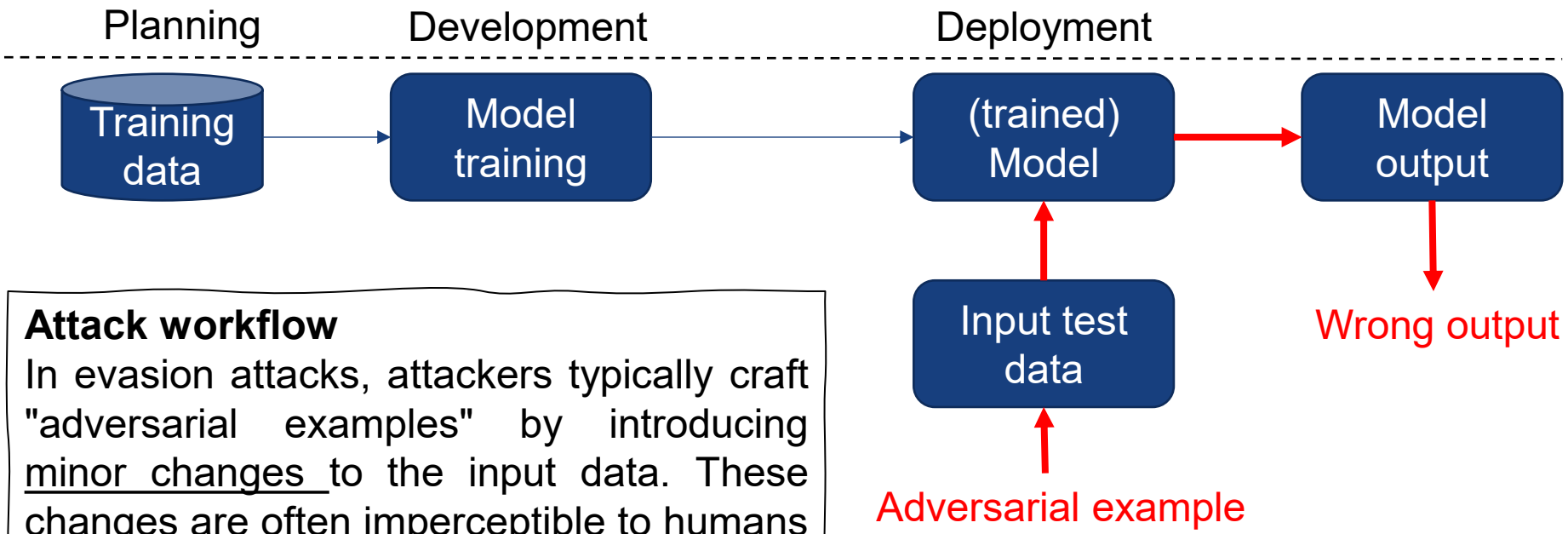
Attack: Evasion attack



Attack workflow

In evasion attacks, attackers typically craft "adversarial examples" by introducing minor changes to the input data. These changes are often imperceptible to humans but can cause significant deviations in the AI model's output. For instance, in image recognition systems, an attacker can add a small amount of noise to an image of a stop sign, making the model classify it as a yield sign or a speed limit sign, which can have dangerous consequences in an autonomous driving scenario.

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Adversarial examples (text data)

Perturbation strategy

- Character-level: Introduce typos or errors to words, e.g., by adding, deleting, repeating, replacing, and permuting characters for certain words.
- Word-level: Replace words with synonyms or contextually similar words.
- Sentence-level: Append irrelevant or extraneous sentences.
- Semantic-level: Simulate linguistic nuances and variations.

Input	Today's class is great, I enjoyed and learned a lot, I know how to apply in my work.
Character-level	T0day's claas is greta, I enjjoyed and lerned a lot, I kno how to aplly in my work.
Word-level	Today's session is excellent, I liked and grasped a lot, I understand how to utilize in my job.
Sentence-level	Today's class is great, I enjoyed and learned a lot, I know how to apply in my work. By the way, the cafeteria was unusually crowded today.
Semantic-level	I found today's class to be quite good; I believe I gained a lot and can potentially apply the concepts in my role.

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Adversarial examples (tabular data)

Perturbation strategy

- Numerical perturbation: Slightly increase or decrease numeric features within reasonable or legal bounds.
- Categorical flipping: Change discrete attributes to those correlated with positive outcomes.


```
{
  "age": 29,
  "annual_income": 45000,
  "job": "bartender",
  "credit_score": 660,
  "owns_home": false,
  "loan_amount": 20000,
  "approved": false
}
```



```
{
  "age": 29,
  "annual_income": 54000,
  "job": "accountant",
  "credit_score": 685,
  "owns_home": true,
  "loan_amount": 18000,
  "approved": true
}
```


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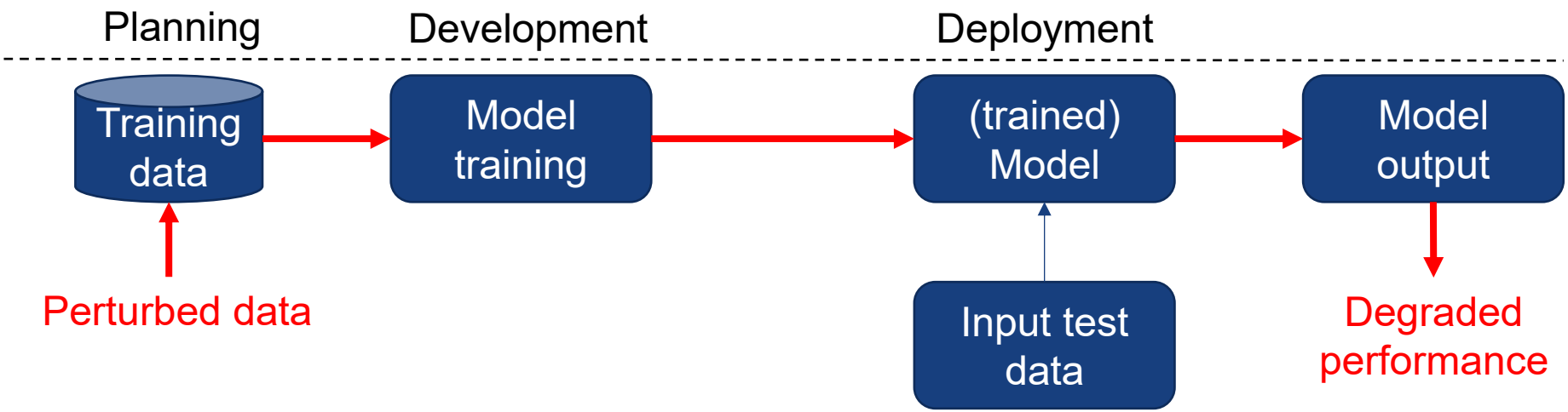
Attack: Poisoning attack



Attack workflow

The attacker introduces perturbed data into the training dataset. These data are crafted to bias the model in ways that may not be immediately noticeable but can significantly degrade performance when the model encounters specific inputs later. For example, in an image classification task, the attacker can apply

- **Label flipping**, which injects training images that are mislabeled, causing the model to associate incorrect labels with particular features;
- **Backdoor poisoning**, which adds a small patch trigger in a subset of images at training time and changes their label to a target class. The classifier learns to associate the trigger with the target class, and any image with the trigger pattern will be misclassified as the target class at testing time.

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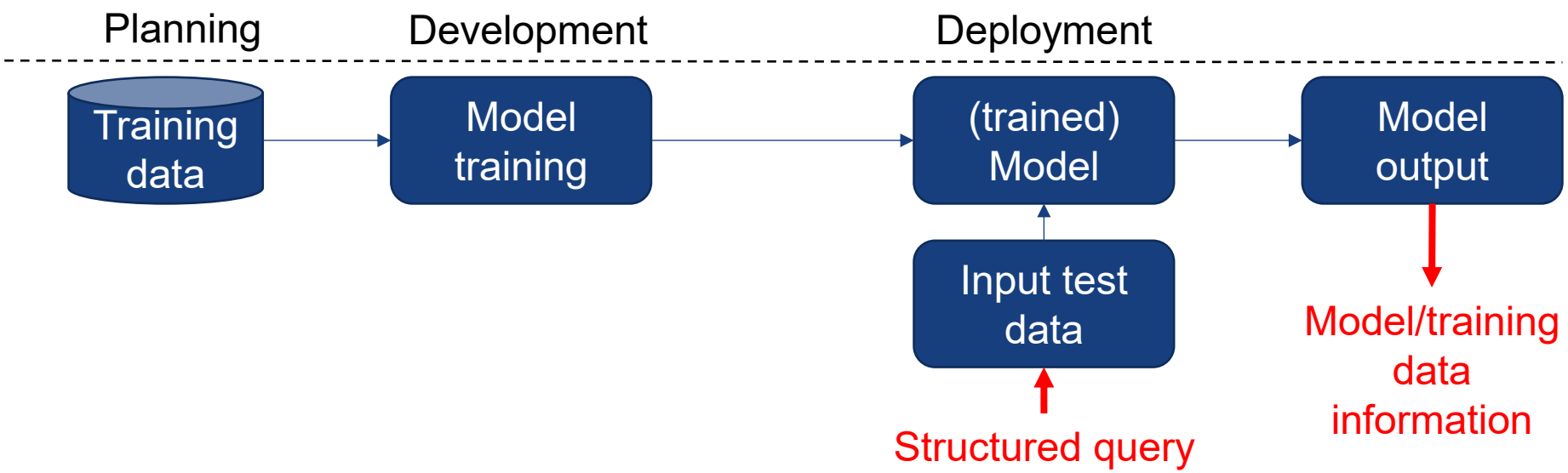
Attack: Privacy attack



Attack workflow

- **Membership inference attack:** the attacker queries the target model with specific input data and analyzes the model's output, such as confidence scores, to determine whether a particular record was part of the training dataset. It exploits the observation that AI models often behave differently on the data that they were trained on versus the data that they "see" for the first time.
- **Model extraction attack:** the attacker submits numerous input queries to the target model, collects the corresponding outputs, and uses this information to reverse-engineer the model's architecture or parameters, ultimately replicating or approximating the original model.

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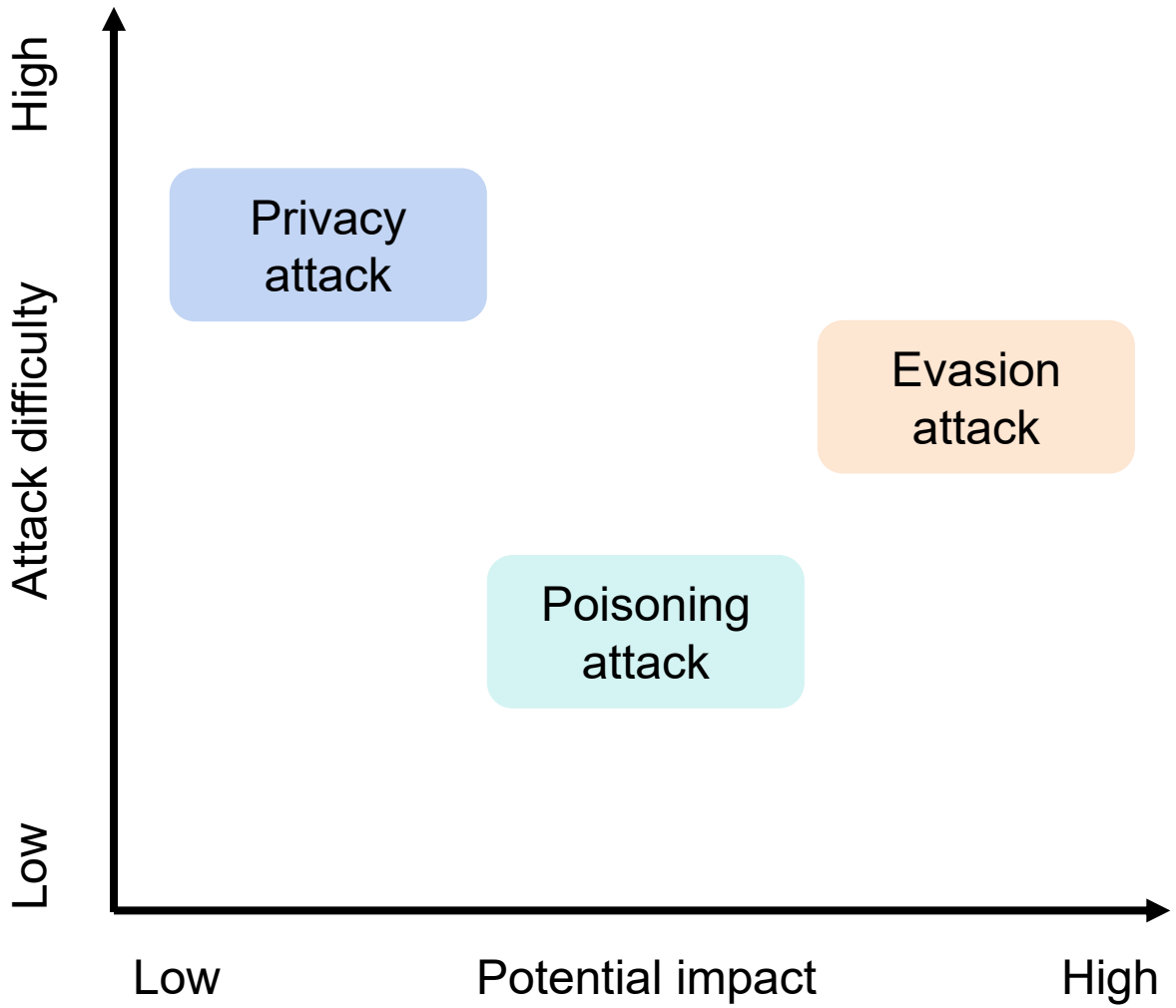
Attack: Summary



Source: https://www.ibm.com/think/insights/mapping-attacks-generative-ai-business-impact

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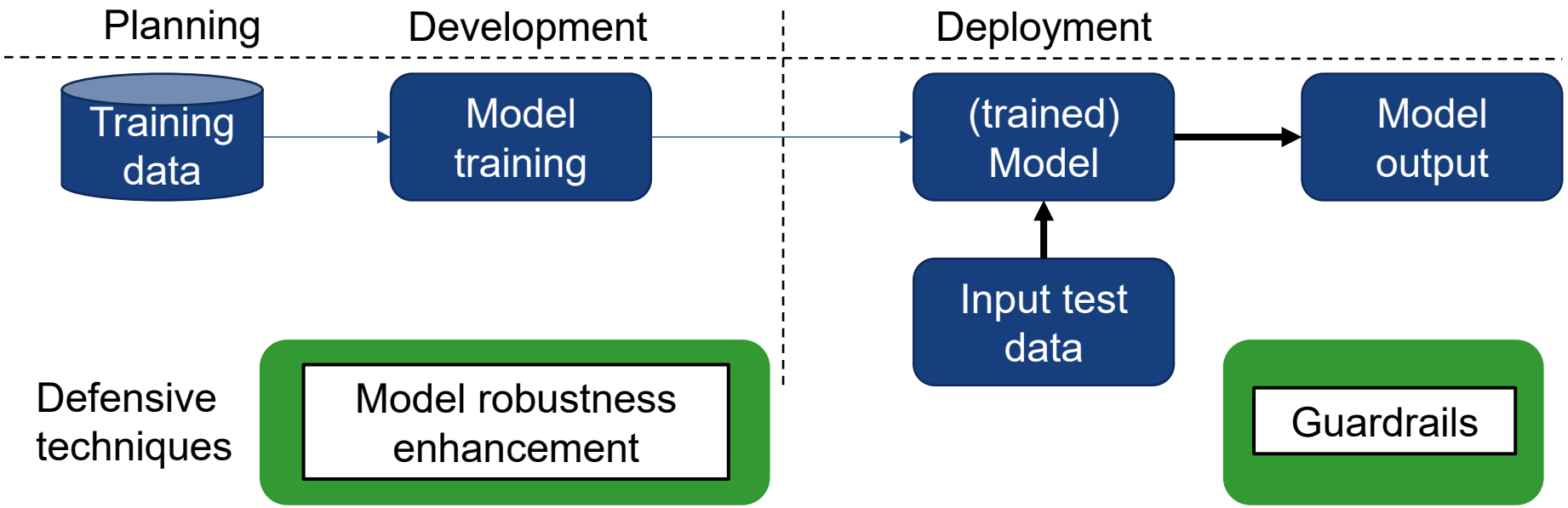
Defensive strategy: Overview



[Model training stage]
Change the machine learning model
- Adversarial re-training
- Ensemble learning

[Model deployment stage]
Without changing the machine learning model
- Input guardrails: Identify adversarial examples in model inference
- Output guardrails: Assess the model's confidence, additional verification for critical decisions.

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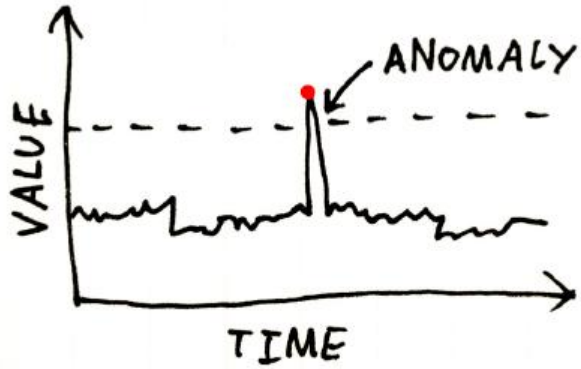
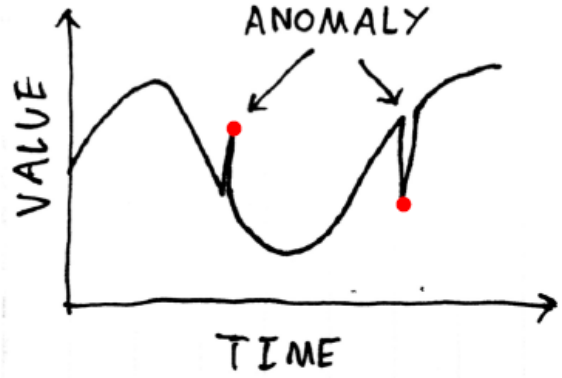
Defensive strategy: Adversarial example detection

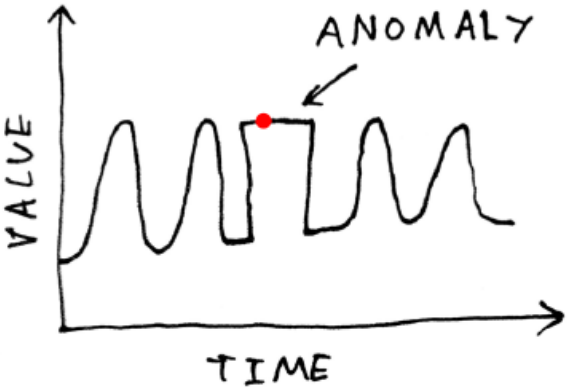
- Idea: Train a machine learning model for anomaly detection, with respect to the training dataset.



Point anomaly



Contextual anomaly



Collective anomaly

Reference: https://www.aisingapore.org/forums/forum-ai-meetups/the-science-of-anomaly-detection-meetup-slides/

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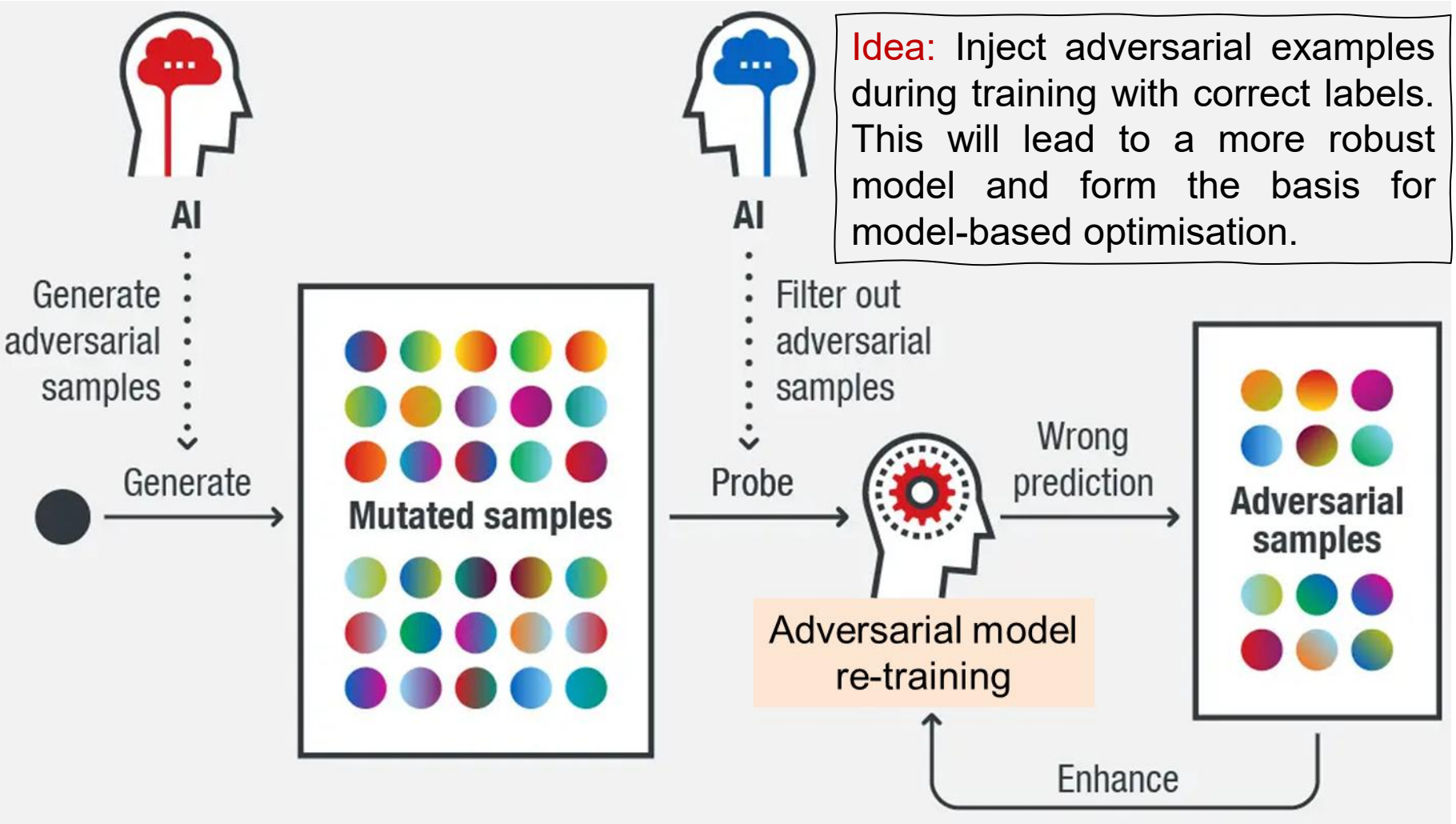
Defensive strategy: Adversarial re-training



Idea: Inject adversarial examples during training with correct labels. This will lead to a more robust model and form the basis for model-based optimisation.

Reference: https://www.trendmicro.com/en_us/research/18/h/adversarial-sample-generation-making-machine-learning-systems-robust-for-security.html

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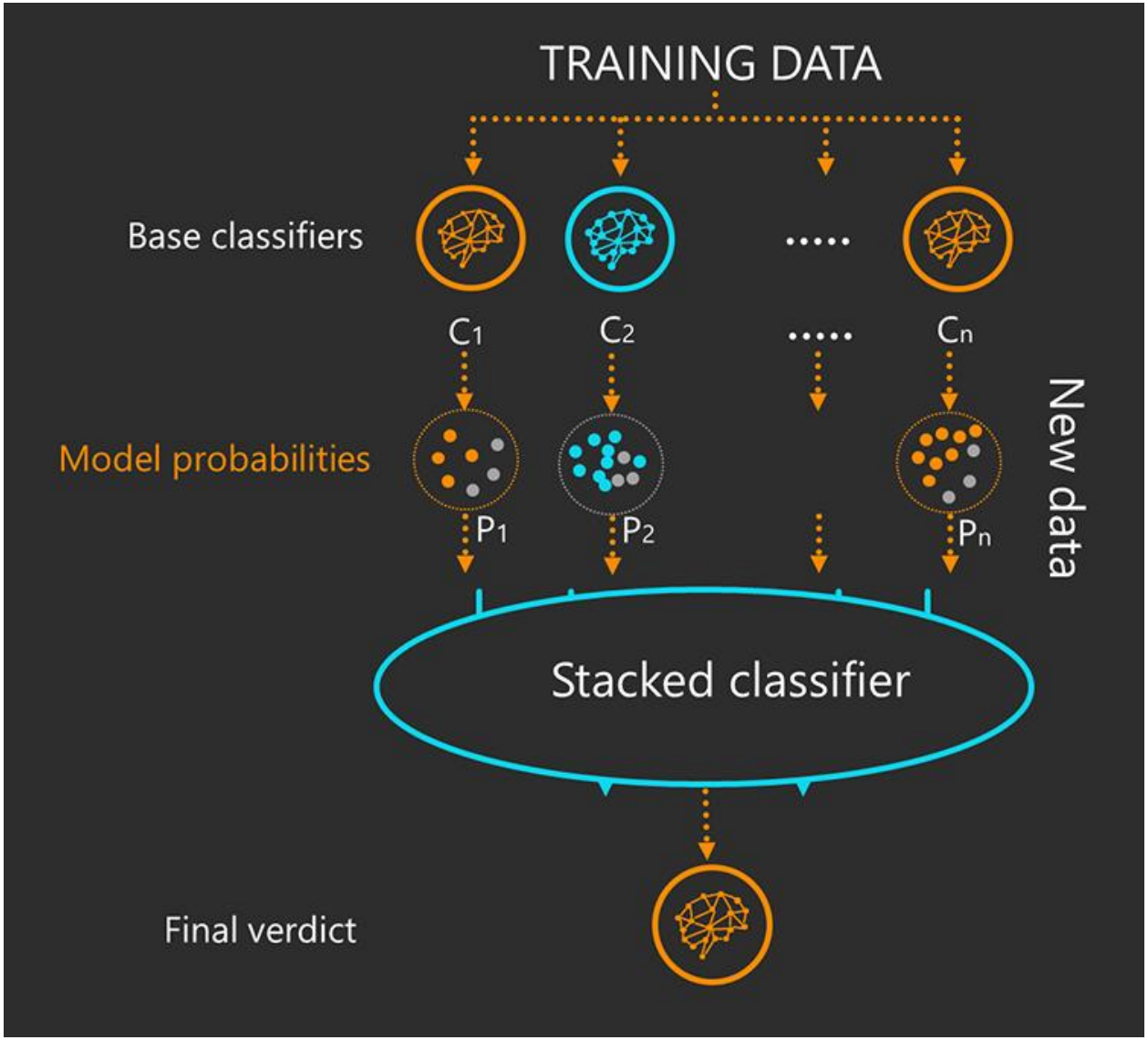
Defensive strategy: Ensemble learning



Reference: https://www.microsoft.com/security/blog/2018/08/09/protecting-the-protector-hardening-machine-learning-defenses-against-adversarial-attacks/

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Testing (toolkit)

Adversarial Robustness Toolbox (ART)	IBM	A Python library for Machine Learning Security. ART provides tools that enable developers and researchers to defend and evaluate Machine Learning models and applications against the adversarial threats of Evasion, Poisoning, Extraction, and Inference. ART supports all popular machine learning frameworks (TensorFlow, Keras, PyTorch, scikit-learn, etc.), all data types (images, tables, audio, video, etc.), and machine learning tasks (classification, object detection, speech recognition, generation, certification, etc.). https://github.com/IBM/adversarial-robustness-toolbox
CleverHans	OpenAI	A Python library developed by Google, designed to benchmark the vulnerability of machine learning models to adversarial examples. It provides tools for generating adversarial inputs and evaluating model robustness. https://github.com/tensorflow/cleverhans
Advbox	Baidu	An adversarial attack toolbox developed by Baidu, providing various attack and defense techniques for AI models, particularly focusing on evasion attacks. https://github.com/advboxes/AdvBox
Foolbox	Bethge lab	A Python toolbox designed by Bethge Lab to assess the robustness of machine learning models. It supports numerous adversarial attacks and offers flexibility in generating adversarial examples across different models and frameworks. https://github.com/bethgelab/foolbox
Counterfit	Microsoft	A tool developed by Microsoft to help security professionals test traditional AI systems for vulnerabilities. It automates adversarial attack simulations, making it easier to evaluate the resilience of machine learning models. https://github.com/Azure/counterfit/

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Testing (toolkit)

Website: https://github.com/aiverify-foundation/aiverify

ROBUSTNESS TOOLBOX

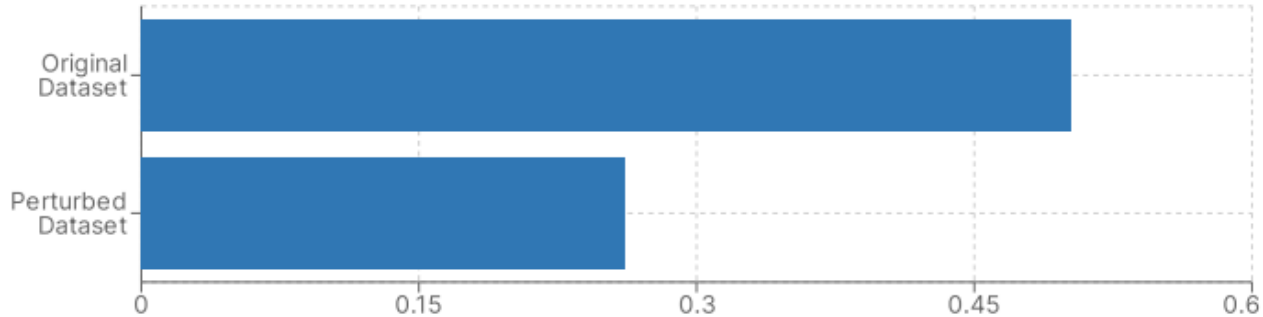
Robustness Toolbox uses Boundary Attack to perturb the test dataset. Boundary Attack is an attack that starts by adding a large amount of noise to a data point intentionally to cause a model it misclassified by the model. We use Salt-and-pepper noise to create the large amount of noise. Then, it will reduce the amount of noise added while maintaining misclassification. This algorithm does not depend on the underlying model's architecture or parameters.

This algorithm is developed for **image dataset** but can also be used to create noise on tabular dataset. However, it is to note that testing on tabular dataset may warrant caution when interpreting the results as this is not well-tested.

Results

Total Number of Samples 2500

Successful Perturbed Rate 100.00%



Each bar represents the performance of the model. The longer the bar, the higher accuracy of the model. A robust model will achieve similar accuracy for both original dataset and perturbed dataset. If you model is not robust, the accuracy of the model will reduce with a perturbed dataset.

What it means:

The test results enable the Company to understand whether the model may be affected by dataset that might be perturbed incidentally or intentionally.

- The original and perturbed dataset achieved an accuracy of 50% and 26% respectively.
- The performance for both datasets are the same.

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Testing (metric)

Vulnerability performance of the AI model

- How the AI model's performance can be affected (or decreased), in terms of commonly-used AI performance indices, such as precision, recall, accuracy, etc.

Robustness performance of the AI model

- The strength of the adversarial examples that can be correctly handled by the AI model to achieve correct results.

Adversarial performance that measures the strength of the adversarial examples.

- It is defined as the percentage of adversarial examples that are successfully misclassified by the target AI model as a wrong target class. A higher score indicates stronger adversarial strength (worse performance of the AI model).

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Testing (red teaming practice)

Planning and preparation

- Deciding to launch the red team
- Identify and allocate budget and resources, and select and contract a third party as needed
- Identifying the target AI system's overview and usage, defining the red teaming scope, scheduling the exercise, and creating a red teaming plan
- Preparing the environment for red teaming

Conducting attacks

- Analyzing risks and developing risk scenarios
- Developing attack scenarios based on the risk scenarios
- Conducting attack scenarios
- Record keeping during red teaming

Reporting and developing improvement plans

- Analyzing the red teaming results
- Preparing the report of results and implementing stakeholder review
- Preparing and reporting the final results
- Developing and implementing improvement plans
- Following up on the progress of the improvement plan and reconducting red teaming as necessary

Reference: Guide to red teaming methodology on AI safety, https://aisi.go.jp/assets/pdf/ai_safety_RT_v1.00_en.pdf

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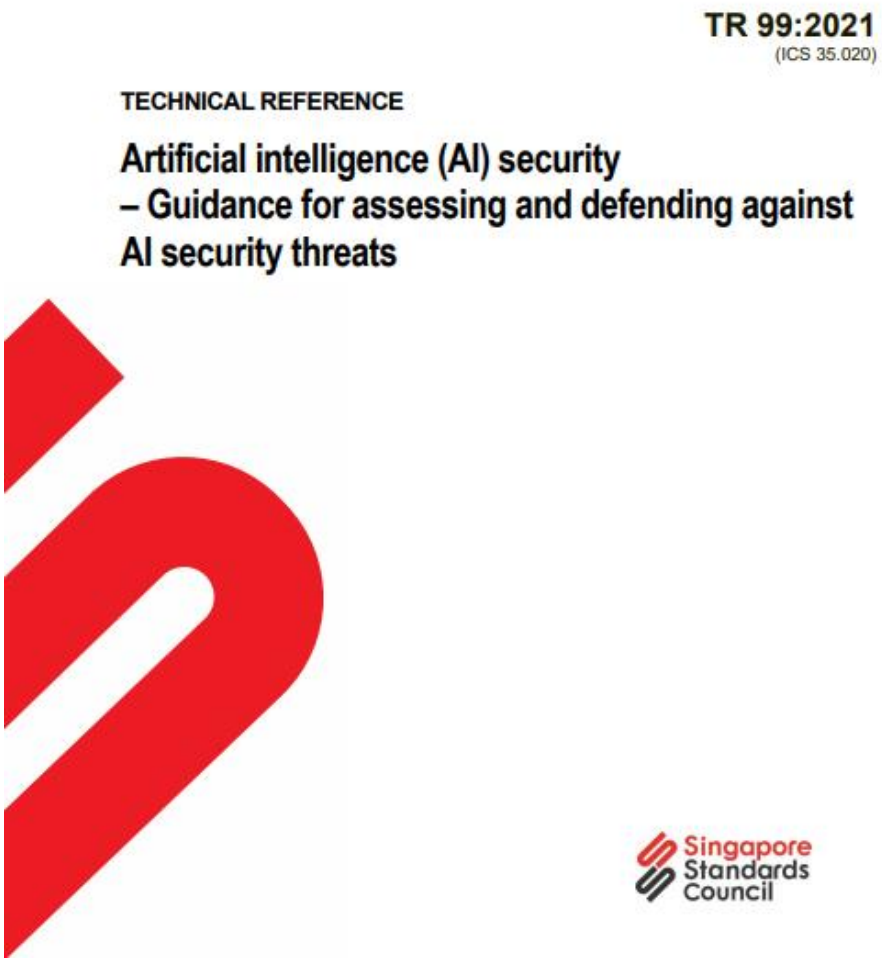
Guidance

TR 99:2021, Artificial intelligence (AI) security – Guidance for assessing and defending against AI security threats


TR 99:2021
(ICS 35.020)

TECHNICAL REFERENCE

Artificial intelligence (AI) security
– Guidance for assessing and defending against
AI security threats

Singapore Standards Council

- An in-depth explanation of AI threats that traditional AI systems can encounter;
- A comprehensive high-level overview of the state-of-the-art security defenses;
- Four use case studies featuring real-world high-risk AI applications aligned with the national AI strategy of Singapore, which serves as validation and a guide for further adoption.
  - Social media – content filtering
  - Finance – credit scoring
  - Healthcare – disease diagnosis
  - Cybersecurity – Malware detection

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Guidance

Model artificial intelligence governance framework (second edition) for traditional AI systems

Singapore launched our Model AI Governance Framework ("Model Framework") at the World Economic Forum in Davos. The Model Framework's unique contribution to the global discourse on AI ethics lies in translating ethical principles into practical recommendations that organisations could readily adopt to deploy AI responsibly.

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Guidance

GUIDELINES ON
SECURING AI SYSTEMS

Guidelines on Securing AI Systems. The Cyber Security Agency of Singapore (CSA) has developed Guidelines on Securing AI Systems for system owners to secure the use of AI throughout its lifecycle. As AI is increasingly integrated into enterprise systems, security should be considered holistically at the system level. All stakeholders involved across the lifecycle of an AI system should seek to better understand the security threats and their potential impact on the desired outcomes of the AI system, and what decisions or trade-offs will need to be made.

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TLDR

Security in machine learning systems is the protection of data, models, and infrastructure from unauthorized access, manipulation, or disruption. It spans the design and implementation of defensive mechanisms that protect against data poisoning, model theft, adversarial manipulation, and system-level vulnerabilities. Security mechanisms ensure the integrity, confidentiality, and availability of machine learning services across development, deployment, and operational environments.

Quoted from Page 1298, Chapter 15, Security & Privacy, Machine Learning Systems, https://www.mlsysbook.ai/

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Thank you!

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